

Dominance of *Methanosarcinales* Phylotypes and Depth-Wise Distribution of Methanogenic Community in Fresh Water Sediments of Sitka Stream from Czech Republic

Prem Prashant Chaudhary · André-Denis G. Wright ·
Lenka Brablcová · Iva Buriánková ·
Adam Bednařík · Martin Rulík

Received: 25 February 2014 / Accepted: 26 May 2014 / Published online: 17 July 2014
© Springer Science+Business Media New York 2014

Abstract The variation in the diversity of methanogens in sediment depths from Sitka stream was studied by constructing a 16S rRNA gene library using methanogen-specific primers and a denaturing gradient gel electrophoresis (DGGE)-based approach. A total of nine different phylotypes from the 16S rRNA library were obtained, and all of them were clustered within the order *Methanosarcinales*. These nine phylotypes likely represent nine new species and at least 5–6 new genera. Similarly, DGGE analysis revealed an increase in the diversity of methanogens with an increase in sediment depth. These results suggest that *Methanosarcinales* phylotypes might be the dominant methanogens in the sediment from Sitka stream, and the diversity of methanogens increases as the depth increases. Results of the present study will help in making effective strategies to monitor the dominant methanogen phylotypes and methane emissions in the environment.

Introduction

One of the most important steps in estimating the global and local carbon accounts is the decomposition of organic

matter in the aquatic sediments as this process recycles the complex organic compounds from fresh water sediments into carbon dioxide and methane [21, 42]. Methane production is considered as an important component of the carbon cycle in these fresh water sediments, and as a result, high methane production has been observed in these environments [21, 42, 44, 53]. Fresh water sediments are the habitats of millions of microorganisms, including bacteria, fungi, protozoa and methanogens, which work together to accomplish biogeochemical processes [16, 47]. Nowadays, much focus has been on studying the diversity of methanogens in these sediments due to their role in global warming.

It has been estimated that freshwater ecosystems, including lakes, wetlands and rice paddies, contribute up to 50 % of the annual atmospheric methane flux [11, 12]. Methane is one of the most potent greenhouse gases that significantly contribute to global warming. Therefore, decreasing the number and activity of methanogens are considered as effective strategies to tackle this problem of global warming [6, 9, 10, 57]. The most common species of methanogens isolated and identified so far from the fresh water sediment samples belong to the genera *Methanosarcina*, *Methanomicrobium*, *Methanobacteriales* and *Methanococcus* [24, 26, 33, 35, 51, 53]. With the advancement of molecular techniques over the last two decades, it is now possible to study the diversity of anaerobic microorganisms in extreme environments, such as deep sediment samples and gut/rumen environments without the need of cultivation [9, 10, 13, 29, 37]. It has become evident that with a change in sediment depth, the gradient of physical and chemical conditions also changes, and provides unique environments for the growth of different types of metabolically diverse microorganisms in the sediment [34, 55].

P. P. Chaudhary (✉) · L. Brablcová · I. Buriánková ·
A. Bednařík · M. Rulík
Laboratory of Aquatic Microbial Ecology, Department of
Ecology and Environmental Sciences, Faculty of Science,
Palacky University, Šlechtitelů 11, 783 71 Olomouc,
Czech Republic
e-mail: luckypgi@gmail.com

A.-D. G. Wright
Department of Animal Science, University of Vermont,
570 Main Street, Burlington, VT 05405, USA

Organic matter (OM), redox potential, dissolved oxygen (DO), temperature and nutrient gradients are the most important factors that can affect the diversity and metabolism of microbial communities inhabiting those environments [8].

In the present study, the molecular diversity of methanogens present in the freshwater sediments, down to a depth of 50 cm, was studied from Sitka stream (Czech Republic) using 16S rRNA gene sequences and a DGGE fingerprinting approach.

Materials and Methods

Study Site

Sampling of the sediments for the study of methanogen community structure was performed from different sampling sites located along Sitka stream in Olomouc, Czech Republic. Five sites along the stream were selected for sampling based upon earlier studies, which showed a considerable amount of methane production and methanogenic potential [3, 40]. This confirmed the suitability of these field sites for the study of diversity of methanogens [21, 22, 40]. Sitka is an undisturbed third order stream with a length of about 35 km. The point of origin of this stream is in the Hrubý Jeseník mountains about 650 m above sea level. Localities I and II are mostly composed of forested area on the upper side, whereas localities III–V are generally considered as an agricultural landscape. In this study, we focused on the overall diversity of methanogens between 0 and 50 cm depth and we examined the methanogenic archaeal community structure at two different depths at locality IV only because this site has the highest methane production [40]. In addition, the physicochemical properties of this locality are well described in one of our earlier studies [4].

Collection and Processing of Sediment Sample

The Freeze-core method, using liquid N₂ as a coolant, was used for the collection of hyporheic sediments [2]. Altogether, three cores were gathered and taken for subsequent analyses. After sampling two layers, the surface (i.e. 0–25 cm) and the remaining layer (i.e. 25–50 cm) were immediately separated for DGGE analysis and stored at low temperature during transport to the laboratory. Samples were then thawed and wet sediment from each layer was sieved, and only particles <1 mm were considered for DNA isolation since most of the microorganisms would be attached to them [31]. Four subsamples (two from each depth) were used for DNA extraction.

DNA Extraction, PCR Amplification and Clone Library Construction

For DNA extraction, 50 g of sediment sample was processed using the PowerSoil DNA Isolation Kit (MO BIO, USA), according to the manufacturer's instructions. DNA was PCR amplified in a BIOER XP thermal Cycler using methanogen specific primers Met86F and Met1340R [56]. The PCR protocol was as follows: 95 °C for 5 min, 39 cycles of 94 °C for 1 min, 60 °C for 1 min, 72 °C for 1.5 min, with a final extension step of 10 min at 72 °C. A 16S rRNA gene library was constructed from the pooled PCR products obtained from each depth (i.e. 0–25 cm and 25–50 cm). Cloning of the 1.2 kb PCR product, selection of positive clones, reamplification of the plasmid DNA and sequencing were carried out using previously published protocols [9, 10]. Sequences of all the primers used in this study are given in Table 1.

BLAST and Statistical Analysis

Sequences were presumptively identified using GenBank's Basic Local Alignment Search Tool (BLAST), and possible chimeras were removed using Chimera Slayer [20]. Good's Coverage, Shannon-Wiener, Simpson, ACE and Chao1 indices were calculated using the bioinformatic suite of programmes called MOTHUR (ver 1.29) [45].

Phylogenetic Analysis

Sequences from valid species, which had the best percent identity, were used as reference sequences and used with the new sequences generated from the current study to construct a phylogenetic tree based upon the neighbour-joining algorithm in MEGA5 [32, 41, 48, 49]. Support for branching was assessed using bootstrap resampling of the sequence data (500 replicates) [14, 49, 50].

Denaturing Gradient Gel Electrophoresis

DGGE was performed using the amplified products from the isolated DNA from both the sediment samples collected from the surface (0–25 cm) and at 25–50 cm depth. Products for DGGE was prepared by PCR amplification of the DNA using specially designed degenerative primers in a 50 µl reaction mixture volume containing the following: 1 µl diethylnitrophenyl thiophosphate (dNTP) mixture (200 µM of each dNTP), 5 µl 10× Taq buffer (with 15 mM MgSO₄), 2 µl of forward primer (200 nM), 2 µl of reverse primer (200 nM), 2 µl of 25 mM MgSO₄, 0.5 µl of Taq DNA Polymerase (Fermentas; 5 U/µl) and 37.5 µl sterile H₂O. The amplified products were cleaned and run on a 45–60 % denaturing gradient gel for 15 h at 85 V. At

Table 1 Primers used for 16S library construction and DGGE analysis

Name	Primer sequence	Annealing temperature (°C)	Reference
Met 86F	5'-GCTCAGTAACACGTGG-3'	60	Wright and Pimm [56]
Met 1340R	5'-CGGTGTGTGCAAGGAG-3'	60	Wright and Pimm [56]
0357 F-GC	5'-CGC CCG CCG CGC GCG GCG GGC GGG GCG GGG GCA CGG GGG G CCC TAC GGG GCG CAG CAG-3'	55	Watanabe et al. [52]
0691 R	5'-GGA TTA CAR GAT TTC AC-3'	55	Watanabe et al. [52]

Table 2 16S rDNA clones from pooled PCR samples from two sediment layers (0–25 cm and 25–50 cm) from locality IV of Sitka stream

16S rDNA gene phylotype	No. of clones	Size (bp)	GenBank accession no.	Nearest valid taxon	%Sequence identity
LAME-D1	10	1250	KF758468	<i>Methanosarcina lacustris</i>	96
LAME-D2	7	1250	KF758469	<i>Methanosarcina vacuolata</i>	96
LAME-D3	4	1250	KF758470	<i>Methanosarcina lacustris</i>	97
LAME-D5	9	1250	KF758472	<i>Methanosarcina vacuolata</i>	97
LAME-D9	12	1250	KF758476	<i>Methanosarcina lacustris</i>	97
LAME-D10	11	1250	KF758477	<i>Methanosarcina barkeri</i>	93
LAME-D11	11	1250	KF758478	<i>Methanosaeta concilii</i>	96
LAME-D12	8	1250	KF758479	<i>Methanosaeta concilii</i>	85
LAME-D14	2	1250	KF758481	<i>Methanosaeta concilii</i>	96

the completion of the run, the gel was stained for 1 h with SYBR Green I nucleic acid gel stain (1:10 000 dilution, Lonza, Rockland, USA). The DGGE gel was then photographed under a ultra-violet illumination (Molecular Dynamics), and the images were analysed using the programme Gel2K (University of Bergen, Norway).

Nucleotide Sequence Accession Numbers

The sequences from this study have been deposited in the GenBank database under the accession numbers from KF758468 to KF758470, KF758472, KF758476 to KF758479, and KF758481.

Results

Microbial Community Composition up to 50 cm Depth in Freshwater Sediment

After the removal of 3 chimeras (LAME-D7, LAME-D8, LAME-D13), a total of 74 sequences were assigned to 9 different phylotypes (LAME-D1, LAME-D2, LAME-D3, LAME-D5, LAME-D9, LAME-D10, LAME-D11, LAME-D12 and LAME-D14) (see Table 2). Phylogenetic analysis showed that the methanogen sequences obtained from this study clustered into one major group. All 74 sequences had a sequence identity between 85 and 97 % to methanogens

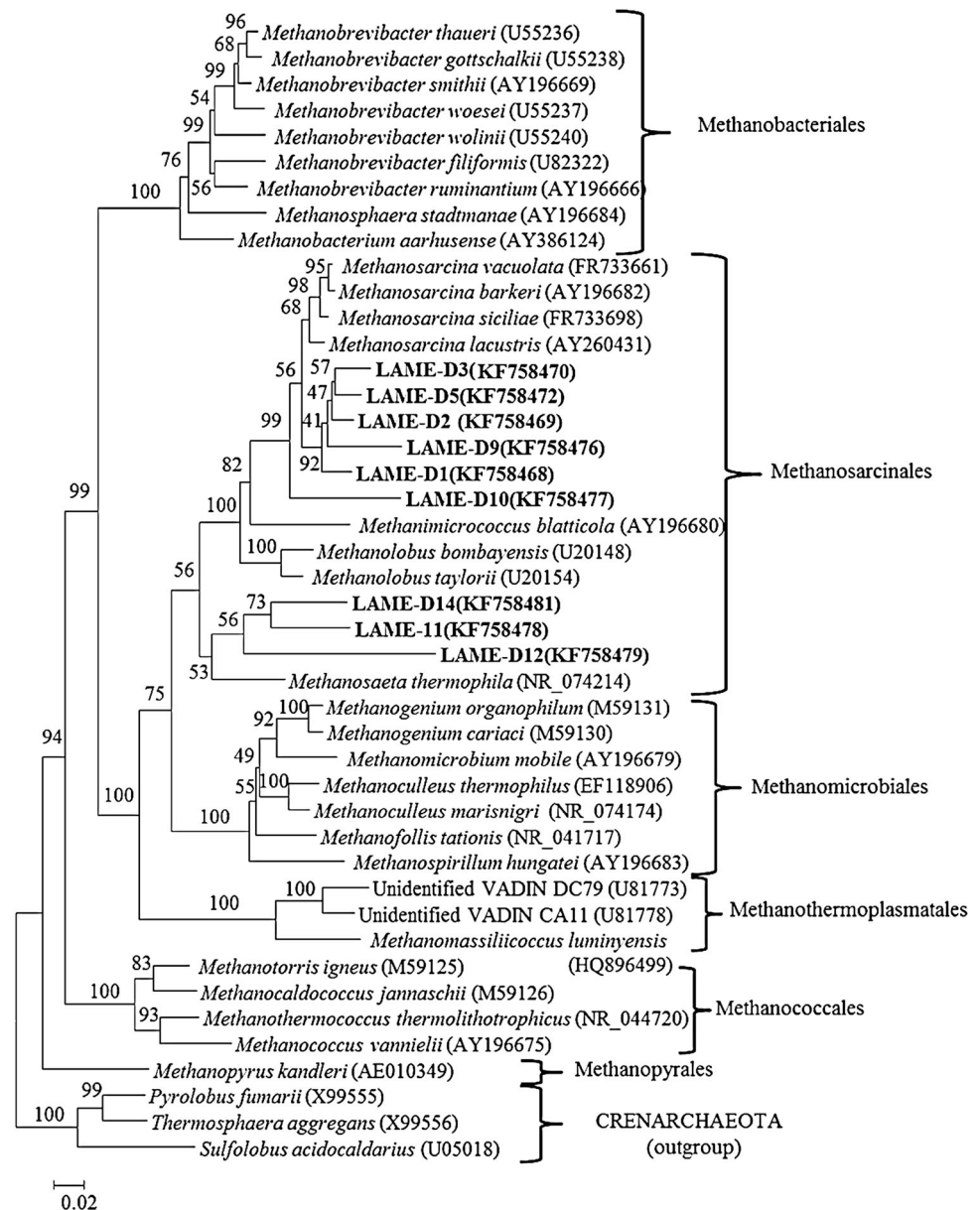
belonging to the order *Methanosarcinales* (Fig. 1). These nine phylotypes likely represent nine new species and at least 5–6 new genera within the order *Methanosarcinales*.

Pairwise distance data (not shown) of these nine phylotypes revealed that the average genetic divergence over all possible pairs of sequences was 18.7 % (variance: 0.0087, std. dev.: 0.0934) with the greatest genetic distance being 33.5 % between clones LAME-D9 and LAME-D12. The minimum distance was 3.8 % between sequences LAME-D1 and LAME-D2. Using a sequence identity criterion of 98 %, MOTHUR assigned the 9 phylotypes to 9 operational taxonomic units (OTUs) [45]. The largest OTU (LAME-D9) comprised 12 clone sequences. None of the 9 OTUs was represented by a single clone sequence.

Two different methods were used to assess the depth of coverage and sampling efficiency at the OTU level. The rarefaction curve approached the saturation point, which was predicted to be 9 OTUs according to the ACE and Chao1 richness indicators (Fig. 2), and Good's coverage was also estimated at 100 %. Together, these results indicate that the library was very well sampled for the diversity that it contained and supports that the sampling efficiency of the current study was very high. Shannon index was 2.107 and the Simpson index was 0.116.

In order to check the vertical distribution of methanogen population, DGGE was performed on methanogen DNA from the two hyporheic sediment depths (locality IV). DGGE revealed a total of 16 different bands (Fig. 3a); 11

Fig. 1 Evolutionary relationships of taxa. The evolutionary history was inferred using the neighbour-joining method [41]. The optimal tree with the sum of branch length = 2.23640570 is shown. The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1,000 replicates) are shown next to the branches [14]. The tree is drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Kimura 2-parameter method [27] and are in the units of the number of base substitutions per site. The analysis involved 44 nucleotide sequences. All positions containing gaps and missing data were eliminated. There were a total of 988 positions in the final dataset. Evolutionary analyses were conducted in MEGA5 [50]



bands in the upper sediment layer (0–25 cm) and 13 DGGE bands in the lower sediment layer (25–50 cm) (Fig. 3b). Eight DGGE bands were detected in both sediment depths, while 3 bands were only found in the upper sediment layer and 4 bands were only found in the deeper sediment layer.

Discussion

The results of the present study were in agreement with the information provided in several recent studies [4, 7]. In the current study, the order *Methanosarcinales* may represent the most dominant group of methanogens in our sediment samples. Chaudhary et al. [7] suggested in their review that

Methanosarcinales and *Methanomicrobiales* phylotypes are the two major phylotypes of methanogens present in the sediment samples all over the world. Grosskopf et al. [19] also reported that *Methanosarcina* spp. was one of the most abundant groups of methanogens in rice roots and in the anoxic bulk soil of flooded rice microcosms.

In another study conducted in the sediment samples in an estuary in the United Kingdom, Purdy et al. [36] reported that most of the clones were closely related to the acetate-utilising methanogen *Methanosaeta concilii*, suggesting that acetate-utilising methanogens are an important component of the methanogen community in these sediments. Purdy et al. [36] also found that the clones similar to *M. concilii* were found to be ubiquitous in freshwater sediments.

A study conducted in Antarctica also found that methanogens belonging to the order *Methanosarcinales* were the most prominent phylotypes in sediment samples

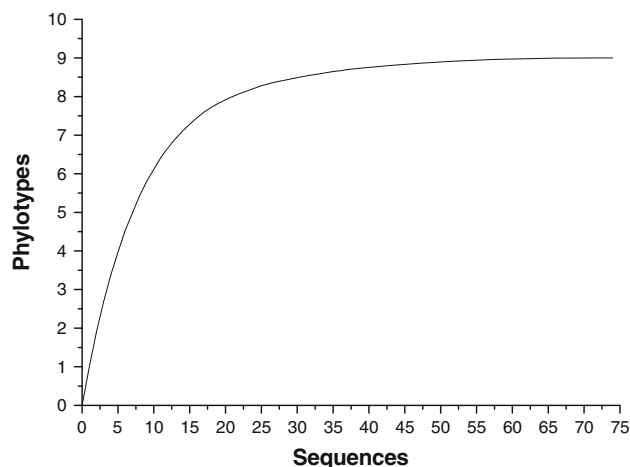


Fig. 2 Collector's rarefaction curve of observed species-level OTUs generated by MOTHUR [45] using a 98 % identity cutoff value

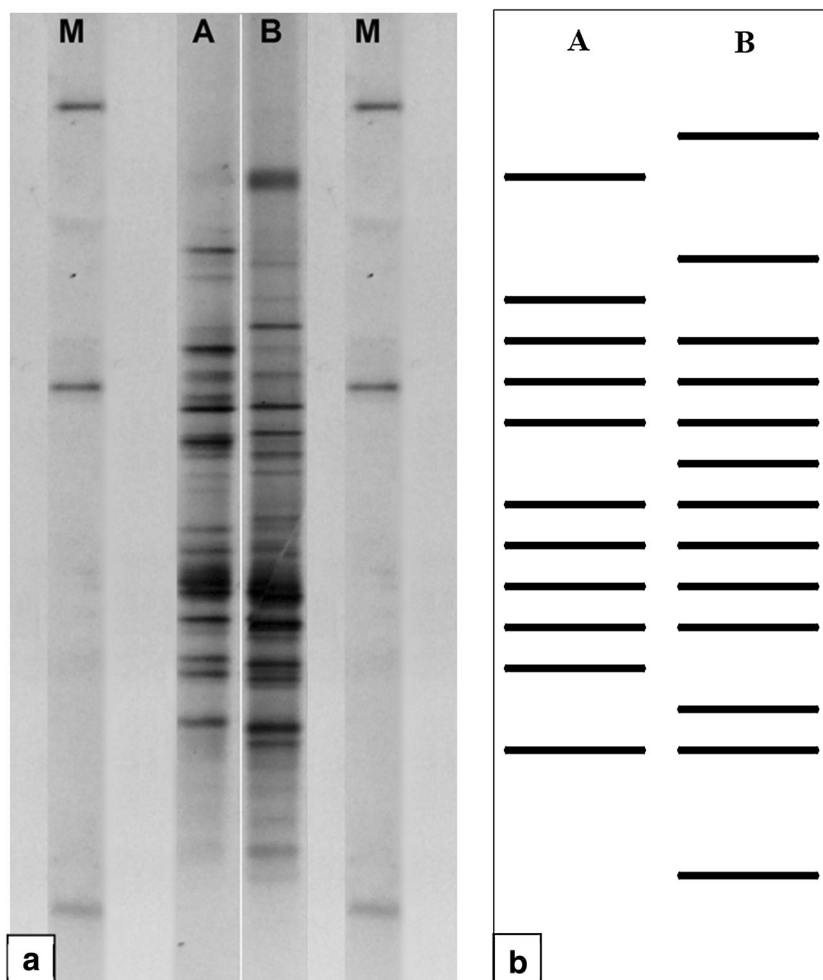
collected from the permanently frozen Lake Fryxell [26]. In a study, Rastogi et al. [38] studied the microbial population in highly metal-contaminated Coeur d'Alene river sediments, but they found just three methanogen phylotypes, all belonging to the order *Methanosarcinales*.

This dominance of *Methanosarcinales* phylotypes may be due to the availability of substrate, such as acetate, in these sediments. In most natural environments, this order of methanogens grows by the breakdown of acetate to methane. In environments where there is no oxygen, lack of light and alternate electron acceptors other than carbon dioxide, acetate utilization is the major source of methane production.

Several other studies also reported the occurrence of *Methanosarcinales* and other phylotypes in sediment samples [1, 25]. Although *Methanobrevibacter* and *Methanomicrobium* phylotypes are the most dominant phylotypes in the rumen of ruminant animals all over the world [9, 10, 17, 58], in the current study, we did not find either of these phylotypes.

Recently one of the review article published by our group gave very detailed information about the major

Fig. 3 DGGE profile (a) and schematic description of the bands was generated via programme Gel2K (University of Bergen, Norway) (b) in upper and deeper sediment layers of studied locality (A upper sediment layer 0–25 cm, B deeper sediment layer 25–50 cm, M molecular marker). **b** Diagrammatical profile represents the no. and position of bands (A upper sediment layer 0–25 cm, B deeper sediment layer 25–50 cm)



phylotypes of methanogens present in sediment samples at different geographical locations [7]. However, a recent paper on methanogen diversity using *mcr-A* gene analysis also suggested that in the Sitka stream, other phylotypes belonging to the orders *Methanomicrobiales*, *Methanobacteriales* and *Methanosarcinales* were also present in abundance [4]. Earlier it was suggested that the members of the order *Methanosarcinales* act as efficient syntrophic partners in the complete degradation of organic biomass in freshwater sediments along with members of order *Methanomicrobiales* [46].

In the current study, the dominance of *Methanosarcinales* phylotype was in agreement with our previous study using the *mcr-A* gene. However, the *mcr-A* gene also showed the presence of *Methanomicrobiales*- and *Methanobacteriales*-related sequences. In the current study, the primers used amplified a much larger portion of the universally conserved 16S rRNA gene whereas *mcr-A* gene primers give the methanogens diversity on the basis of *mcr-A* gene sequence variations. So these changes in the diversity of methanogens might be due to the difference in the specificity and targets of the primers used. A possible explanation for the dominance of *Methanosarcinales* phylotype might be the occurrence of high concentrations of acetate and lactate in the interstitial water even in oligotrophic streams, which will act as a substrate for anaerobic fermentation [39]. *Methanosarcina* and *Methanosaeta* are the only group of methanogens capable of utilizing acetate [15, 18]. The acetate concentrations in interstitial water ranged from 0.23 to 5.25 mg l⁻¹ (0.004–0.087 mM l⁻¹) and tended to increase with sediment depth (Bednářik et al., in review). This acetate concentration does not seem to be sufficient for *Methanosarcina* spp. as they have a threshold of 0.2–1.2 mM acetate, while *Methanosaeta* species require <10 µM acetate. Although the *Methanosarcinales* phylotypes in the current study are the dominant acetoclastic methanogens in the Sitka hyporheic sediments, they may also use H₂-CO₂ and methyl compounds rather than acetate. Similar acetate concentrations have been reported by Sanz et al. [43] who found acetate concentration in the range from 0.3 to 79 mg l⁻¹ in sediments of Rio Tinto, an extreme acidic river.

Our results on the methanogenic community structure are in accordance with our concurrent findings that both acetate and H₂-CO₂ might serve as methanogenic precursors. Based on isotopic composition of produced methane and its precursors, we have found that acetoclastic methanogenesis was either predominant or as prevalent as the hydrogenotrophic methanogenesis in both methanogenic maxima occurring in the hyporheic sediment depth of 40–50 cm (unpublished data). In their study of Lake Dugow sediment, Chan et al. [5] found that the relative

abundance of *Methanoseaceae* decreased with depth, whereas that of *Methanomicrobiales* slightly increases. The structure of the community consisting of acetotrophic (*Methanoseaceae*) versus hydrogenotrophic (*Methanomicrobiales*) phenotypes reflected the vertical distribution of functional characteristics (CH₄ production from acetate versus H₂/CO₂).

The results of the DGGE analysis indicated changes in the methanogenic community structure between the two sediment samples taken at different depths. The number of different methanogen DGGE bands in the present study was compared with observations from paddy field soils in Japan [24, 30, 52]. It appears that the diversity of methanogenic archaea increases as you go deeper down in the sediment. Indeed, we found more DGGE bands in the lower sediment layer than in the upper sediment layer. Similar findings are reported by Huang and his co-workers [23] in China where they found that the number of bacterial bands was higher with increasing sediment depth in Pearl River. Similarly, Koch et al. [28] who detected a higher number of methanogens in deeper sediment layers of submarine permafrost in the Siberian Laptev Sea. However, Xingqing et al. [59] revealed an inverse relationship between the number of bands and increasing sediment depth in Lake Tahu (China).

The present findings on the methanogen composition and diversity in the sediments from Sitka stream demonstrated that *Methanosarcinales* phylotypes appear to be the dominant methanogens and that the diversity of methanogens slightly increased as the sediment depth increased. This might be explained as follows: as the depth increases, the environment becomes more anoxic/anaerobic, which will support the growth of these diverse groups of methanogens. However, a more detailed study at different times of the year is required to check the overall methanogen diversity and shift in the diversity with increased sediment depth. It could also be due to nutrient availability. A detailed study of the geochemical cycle may give a better understanding of this fact. Future work will also utilize Illumina MiSeq ver 3 technology (2 × 300 paired ends) to generate greater sequence coverage.

Acknowledgments The authors are thankful to the European Social Fund and state budget of the Czech Republic for providing the financial support during this study. This work is a part of the POSTUP II project CZ.1.07/2.3.00/30.0041, which is mutually financed by the previously stated funding agencies.

References

1. Biderre-Petit C, Jézéquel D, Dugat-Bony E, Lopes F, Kuever J, Borrel G et al (2011) Identification of microbial communities involved in the methane cycle of a freshwater meromictic lake. *FEMS Microbiol Ecol* 77:533–545

2. Bretschko G, Klemens WE (1986) Quantitative methods and aspects in the study of the interstitial fauna of running waters. *Stygologia* 2:297–316
3. Buriánková I, Brablcová L, Mach V, Hýblová A, Badurová P, Cupalová J et al (2012) Methanogens and methanotrophs distribution in the hyporheic sediments of a small lowland stream. *Fund Appl Limnol* 181:87–102
4. Buriánková I, Brablcová L, Mach V, Dvořák P, Chaudhary PP, Rulík M (2013) Identification of methanogenic archaea in the Hyporheic Sediment of Sitka Stream. *PLoS ONE* 8(11):e80804. doi:10.1371/journal.pone.0080804
5. Chan OC, Claus P, Casper P, Ulrich A, Lueders T, Conrad R (2005) Vertical distribution of structure and function of the methanogenic archaeal community in Lake Dagow sediment. *Environ Microbiol* 7:1139–1149
6. Chaudhary PP (2009) *Methanomicrobium* phylotype are the dominant methanogen phylotype in the Murrah buffaloes. *Lett Appl Microbiol* 48:386
7. Chaudhary PP, Brablcová L, Buriánková I, Rulík M (2013) Molecular diversity and tools for deciphering the methanogen community structure and diversity in freshwater sediments. *Appl Microbiol Biotechnol* 97:7553–7562
8. Chunleuchanon S, Sooksawang A, Teaumroong N, Boonkerd N (2003) Diversity of nitrogen-fixing cyanobacteria under various ecosystems of Thailand: population dynamics as affected by environmental factors. *World J Microbiol Biotechnol* 19:167–173
9. Chaudhary PP, Sirohi SK, Saxena J (2012) Diversity analysis of methanogens in rumen of *Bubalus bubalis* by 16S riboprinting and sequence analysis. *Gene* 493:13–17
10. Chaudhary PP, Sirohi SK (2009) Dominance of *Methanomicrobium* phylotype in rumen (*Bubalus bubalis*) methanogens from India. *Lett Appl Microbiol* 49:274–277
11. Cicerone RJ, Oremland RS (1988) Biogeochemical aspects of atmospheric methane. *Global Biogeochem Cy* 1:61–86
12. Conrad R (2009) The global methane cycle: recent advances in understanding the microbial processes involved. *Environ Microbiol Rep* 1:285–292
13. Dhillon A, Lever M, Lloyd KG, Albert DB, Sogin ML, Teske A (2005) Methanogen diversity evidenced by molecular characterization of methyl coenzyme M reductase A (*mcrA*) genes in hydrothermal sediments of the Guaymas Basin. *Appl Environ Microbiol* 71:4592–4601
14. Felsenstein J (1985) Confidence limits on phylogenies: an approach using the bootstrap. *Evolution* 39:783–791
15. Ferry JG (1993) *Methanogenesis: ecology, physiology, biochemistry & genetics*. Chapman & Hall, New York, 536 pp
16. Fortuna A (2012) The Soil Biota. *Nature Education Knowledge* 3:1
17. Franzolin R, St-Pierre B, Northwood K, Wright ADG (2012) Analysis of rumen methanogen diversity in water buffaloes (*Bubalus bubalis*) under three different diets. *Microb Ecol* 64:131–139
18. Garcia JL, Patel BKC, Ollivier B (2000) Taxonomic, phylogenetic, and ecological diversity of methanogenic archaea. *Anaerobe* 6:205–226
19. Grosskopf R, Stubner S, Liesack W (1998) Novel euryarchaeotal lineages detected on rice roots and in the anoxic bulk soil of flooded rice microcosms. *Appl Environ Microb* 64:4983–4989
20. Haas BJ, Gevers D, Earl AM, Feldgarden M, Ward DV, Gianoukos G et al (2011) Chimeric 16S rRNA sequence formation and detection in Sanger and 454-pyrosequenced PCR amplicons. *Genome Res* 21:494–504
21. Hlaváčová E, Rulík M, Čáp L (2005) Anaerobic microbial metabolism in hyporheic sediment of a gravel bar in a small lowland stream. *River Res Appl* 21:1003–1011
22. Hlaváčová E, Rulík M, Čáp L, Mach V (2006) Greenhouse gases (CO₂, CH₄, N₂O) emissions to the atmosphere from a small lowland stream. *Arch Hydrobiol* 165:339–353
23. Huang S, Chen C, Wu Y, Wu Q, Zhang R (2011) Characterization of depth-related bacterial communities and their relationships with the environmental factors in the river sediments. *World J Microbiol Biotechnol* 27:2655–2664
24. Ikenaga M, Asakawa S, Muraoka Y, Kimura M (2004) Methanogenic archaeal communities in rice roots grown in flooded soil pots: Estimation by PCR-DGGE and sequence analyses. *Soil Sci Plant Nutr* 50:701–711
25. Jiang L, Zheng Y, Chen J, Xiao X, Wang F (2011) Stratification of Archaeal communities in shallow sediments of the Pearl River Estuary, Southern China. *Anton Van Leeuwenhoek* 99:739–751
26. Karr EA, Ng JM, Belchik SM, Sattley WM, Madigan MT, Achenbach LA (2006) Biodiversity of methanogenic and other archaea in the permanently frozen Lake Fryxell, Antarctica. *Appl Environ Microbiol* 72:1663–1666
27. Kimura M (1980) A simple method for estimating evolutionary rate of base substitutions through comparative studies of nucleotide sequences. *J Mol Evol* 16:111–120
28. Koch K, Knoblauch C, Wagner D (2009) Methanogenic community composition and anaerobic carbon turnover in submarine permafrost sediments of the Siberian Laptev Sea. *Environ Microbiol* 11(3):657–668
29. Lazar CS, John Parkes R, Cragg BA, L'Haridon S, Toffin L (2012) Methanogenic activity and diversity in the centre of the Amsterdam Mud Volcano, Eastern Mediterranean Sea. *FEMS Microbiol Ecol* 81:243–254
30. Lee GC, Watanabe T, Murase J, Asakawa S, Kimura M (2012) Growth of methanogens in an oxic soil microcosm: Elucidation by a DNA-SIP experiment using ¹³C-labeled dried rice callus. *Appl Soil Ecol* 58:37–44
31. Leichtfried M (1988) Bacterial substrates in gravel beds of a second order alpine stream (Project Ritrodat-Lunz, Austria). *Verh Internat Verein Limnol* 23:1325–1332
32. Madden TL, Tatusov RL, Zhang J (1996) Applications of network BLAST server. In: Russell, F.D. (Ed.), *Method Enzymol Academic Press*:131–141
33. Munson MA, Nedwell DB, Embley TM (1997) Phylogenetic diversity of Archaea in sediment samples from a coastal salt marsh. *Appl Environ Microbiol* 63:4729–4733
34. Newberry CJ, Webster G, Cragg BA, Parkes RJ, Weightman AJ, Fry JC (2004) Diversity of prokaryotes and methanogenesis in deep subsurface sediments from the Nankai Trough, Ocean Drilling Program Leg 190. *Environ Microbiol* 6(3):274–287
35. Orphan VJ, Jahnke LL, Embaye T, Turk KA, Pernthaler A, Summons RE et al (2008) Characterization and spatial distribution of methanogens and methanogenic biosignatures in hypersaline microbial mats of Baja California. *Geobiology* 6:376–393
36. Purdy KJ, Munson MA, Nedwell DB, Martin Embley T (2002) Comparison of the molecular diversity of the methanogenic community at the brackish and marine ends of a UK estuary. *FEMS Microb Ecol* 39:17–21
37. Putkinen A, Juottonen H, Juutinen S, Tuittila ES, Fritze H, Yrjälä K (2009) Archaeal rRNA diversity and methane production in deep boreal peat. *FEMS Microbiol Ecol* 70:87–98
38. Rastogi G, Barua S, Sani RK, Peyton BM (2011) Investigation of microbial populations in the extremely metal-contaminated Coeur d'Alene river sediments. *Microb Ecol* 62:1–13
39. Rulík M, Čáp L, Hlaváčová E (2000) Methane in the hyporheic zone of a small lowland stream (Sitka, Czech Republic). *Limnologica* 30:359–366
40. Rulík M, Bednarik A, Mach V, Brablcová L, Buriánková I, Badurova P et al (2013) Methanogenic system of a small lowland

- stream Sitka, Czech Republic. Biomass Now-Cultivation Utilization Chapter 17:395–426
41. Saitou N, Nei M (1987) The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Mol Biol Evol* 4:406–425
 42. Sanders IA, Heppell CM, Cotton JA, Wharton G, Hildrew AG, Flowers EJ et al (2007) Emissions of methane from chalk streams has potential implications for agricultural practices. *Freshwater Biol* 52:1176–1186
 43. Sanz JL, Rodríguez N, Díaz EE, Amils R (2011) Methanogenesis in the sediments of Rio Tinto, an extreme acidic river. *Environ Microbiol* 13:2336–2341
 44. Schindler JE, Krabbenhoft DP (1998) The hyporheic zone as a source of dissolved organic carbon and carbon gases to a temperate forested stream. *Biogeochemistry* 43:157–174
 45. Schloss PD, Westcott SL, Ryabin T, Hall JR, Hartmann M, Hollister EB et al (2009) Introducing MOTHUR: open-source, platform-independent, community-supported software for describing and comparing microbial communities. *Appl Environ Microbiol* 75:7537–7541
 46. Schwarz JIK, Eckert W, Conrad R (2007) Community structure of Archaea and Bacteria in a profundal lake sediment Lake Kinneret (Israel). *Syst Appl Microbiol* 30:239–254
 47. Surakasi VP, Wani AA, Shouche YS, Ranade DR (2007) Phylogenetic analysis of methanogenic enrichment cultures obtained from Lonar Lake in India: isolation of *Methanocalculus* sp. and *Methanoculleus* sp. *Microb Ecol* 54:697–704
 48. Takezaki N, Rzhetsky A, Nei M (2004) Phylogenetic test of the molecular clock and linearized trees. *Mol Biol Evol* 12:823–833
 49. Tamura KNM, Kumar S (2004) Prospects for inferring very large phylogenies by using the neighbor-joining method. *Proc Natl Acad Sci USA* 101:11030–11035
 50. Tamura K, Peterson D, Peterson N, Stecher G, Nei M, Kumar S (2011) MEGA5: Molecular evolutionary genetics analysis using maximum likelihood, evolutionary distance, and maximum parsimony methods. *Mol Bio Evol* 28:2731–2739
 51. Watanabe T, Asakawa S, Nakamura A, Nagaoka K, Kimura M (2004) DGGE method for analyzing 16S rDNA of methanogenic archaeal community in paddy field soil. *FEMS Microbiol Lett* 232:153–163
 52. Watanabe T, Hosen Y, Agbisit R, Llorca L, Fujita D, Asakawa S et al (2010) Changes in community structure and transcriptional activity of methanogenic archaea in a paddy field soil brought about by a water-saving practice—Estimation by PCR-DGGE and qPCR of 16S rDNA and 16S rRNA. 19th World Congress of Soil Solutions for a Changing World. Brisbane, Australia, pp 5–8
 53. Weber S, Lueders T, Friedrich MW, Conrad R (2001) Methanogenic populations involved in the degradation of rice straw in anoxic paddy soil. *FEMS Microbiol Ecol* 38:11–20
 54. Wilcock RJ, Sorrell BK (2008) Emissions of greenhouse gases CH₄ and N₂O from low gradient streams in agriculturally developed catchments. *Water Air Soil Poll* 188:155–170
 55. Wilms R, Kopke B, Sass H, Chang TS, Cypionka H, Engelen B (2006) Deep biosphere-related bacteria within the subsurface of tidal flat sediments. *Environ Microbiol* 8:709–719
 56. Wright ADG, Pimm C (2003) Improved strategy for presumptive identification of methanogens using 16S riboprinting. *J Microbiol Methods* 55:337–349
 57. Wright ADG, Auckland CH, Lynn DH (2007) Molecular diversity of methanogens in feedlot cattle from Ontario and Prince Edward Island, Canada. *Appl Environ Microbiol* 73:4206–4210
 58. Wright ADG, Ma X, Obispo NE (2008) *Methanobrevibacter* phylotypes are the dominant methanogens in sheep from Venezuela. *Microb Ecol* 56:390–394
 59. Xingqing Z, Liuyan Y, Zhenyang Y, Naiying P, Lin X, Daqiang Y et al (2008) Characterization of depth-related microbial communities in lake sediment by denaturing gradient gel electrophoresis of amplified 16S rRNA fragments. *J Environ Sci* 20:224–230